


```

main.py 2 database.py X styles.py
OK > database.py > Database > load
4 class Database:
29     def initialize(self):
48         ]
49     else:
50         default = []
51
52         os.makedirs(os.path.dir
53         self.save(key, default)
54
55     def load(self, key):
56         with open(self.files[key], 'r')
57         return json.load(f)
58
59     def save(self, key, data):
60         with open(self.files[key], 'w')
61         json.dump(data, f, indent=4
62

```

```

4 class Database:
29     def initialize(self):
48         ]
49     else:
50         default = []
51
52         os.makedirs(os.path.dirname(path), exist_ok=True)
53         self.save(key, default)
54
55     def load(self, key):
56         with open(self.files[key], 'r') as f:
57             return json.load(f)
58
59     def save(self, key, data):
60         with open(self.files[key], 'w') as f:
61             json.dump(data, f, indent=4)
62

```


File

Edit

Selection

View

Go

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Terminal

Help

Quiz_Project

EXPLORER

OPEN EDITORS

QUIZ_PROJECT

OK

__pycache__

database.py

gui_components.py

main.py

manager.py

pdf_parser.py

stats_engine.py

styles.py

questions.json

results.json

users.json

main.py 2

stats_engine.py

styles.py

OK > stats_engine.py > ...

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import numpy as np

class StatsEngine:

@staticmethod

def calculate_metrics(results, user=None):

if user:

user_results = [r for r in results if r.get('name') == user]

else:

user_results = results

if not user_results:

return None

scores = np.array([r['score'] for r in user_results])

totals = np.array([r['total'] for r in user_results])

percentages = (scores / totals) * 100

return {

"total_attempts": len(scores),

"average_score": float(np.mean(scores)),

"max_score": int(np.max(scores)),

"min_score": int(np.min(scores)),

"median_score": float(np.median(scores)),

"std_deviation": float(np.std(scores)),

"pass_percentage": float((np.sum(percentages >= 50.0) / len(percentages)) * 100)

}

CHAT

SESSIONS

Tip: Try the Plan agent to research and plan before implementing changes.

+ stats_engine.py

Describe what to build

+ Agent Auto

OUTLINE

TIMELINE

FileEditSelectionViewGoRunTerminalHelp

Quiz_Project

main.py 2stats_engine.py Xstyles.py

EXPLORER

OPEN EDITORS

QUIZ_PROJECT

OK

__pycache__

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questions.json

results.json

users.json

stats_engine.py > ...

1import numpy as np

2

3class StatsEngine:

4 @staticmethod

5 def calculate_metrics(results, user=None):

6 if user:

7 user_results = [r for r in results if r.get('name') == user]

8 else:

9 user_results = results

10

11 if not user_results:

12 return None

13

14 scores = np.array([r['score'] for r in user_results])

15 totals = np.array([r['total'] for r in user_results])

16 percentages = (scores / totals) * 100

17

18 return {

19 "total_attempts": len(scores),

20 "average_score": float(np.mean(scores)),

21 "max_score": int(np.max(scores)),

22 "min_score": int(np.min(scores)),

23 "median_score": float(np.median(scores)),

24 "std_deviation": float(np.std(scores)),

25 "pass_percentage": float((np.sum(percentages >= 50.0) / len(percentages)) * 100)

26 }

27

CHAT

SESSIONS

Tip: Try the Plan agent to research and plan before implementing changes.

+ stats_engine.py

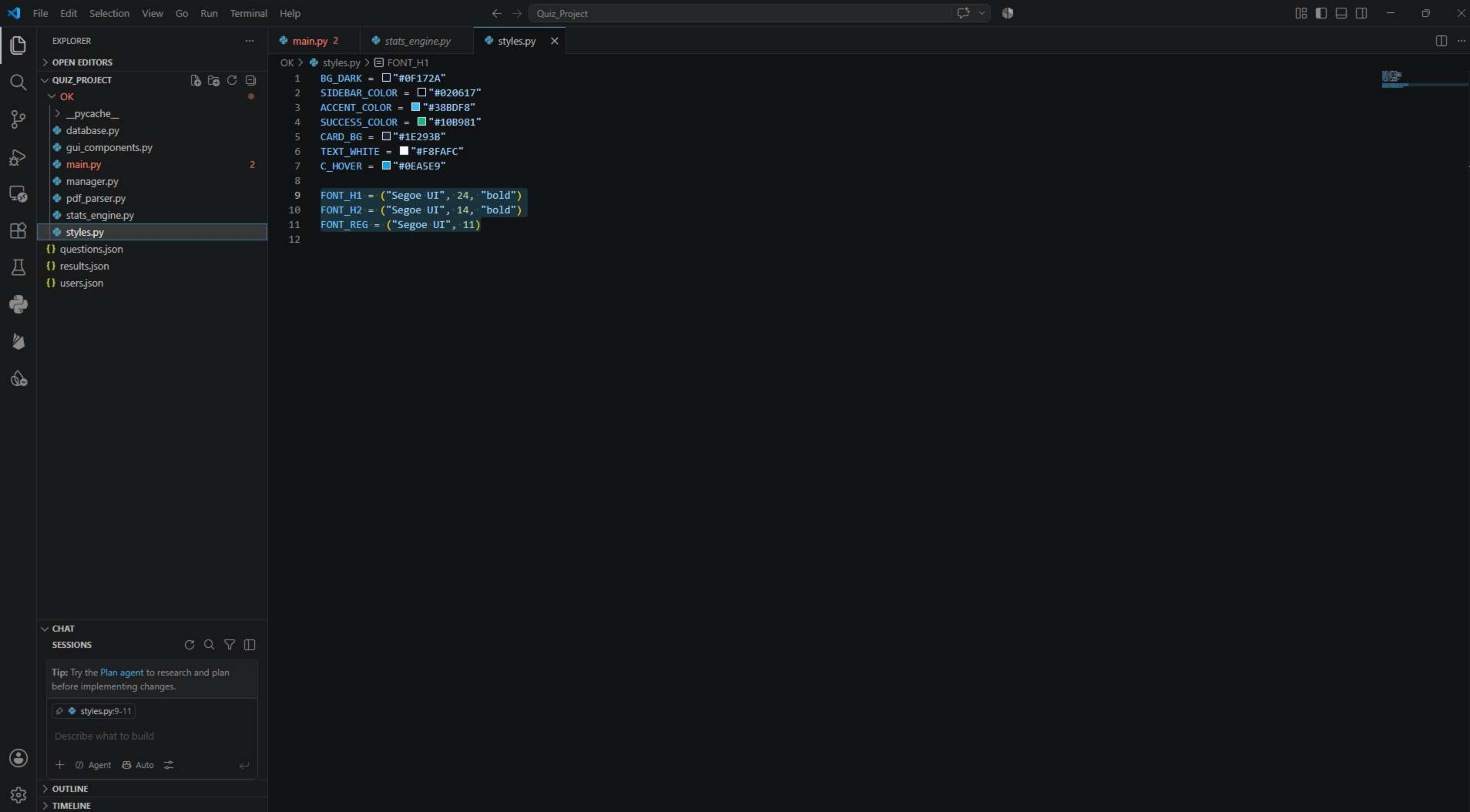
Describe what to build

+ Agent

Auto

OUTLINE

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users.json

main.py 2

stats_engine.py

styles.py X

OK > styles.py > FONT_H1

1 BG_DARK = "#0F172A"

2 SIDEBAR_COLOR = "#020617"

3 ACCENT_COLOR = "#38BDF8"

4 SUCCESS_COLOR = "#10B981"

5 CARD_BG = "#1E293B"

6 TEXT_WHITE = "#F8FAFC"

7 C_HOVER = "#0EA5E9"

8

9 FONT_H1 = ("Segoe UI", 24, "bold")

10 FONT_H2 = ("Segoe UI", 14, "bold")

11 FONT_REG = ("Segoe UI", 11)

12

CHAT

SESSIONS

Tip: Try the Plan agent to research and plan before implementing changes.

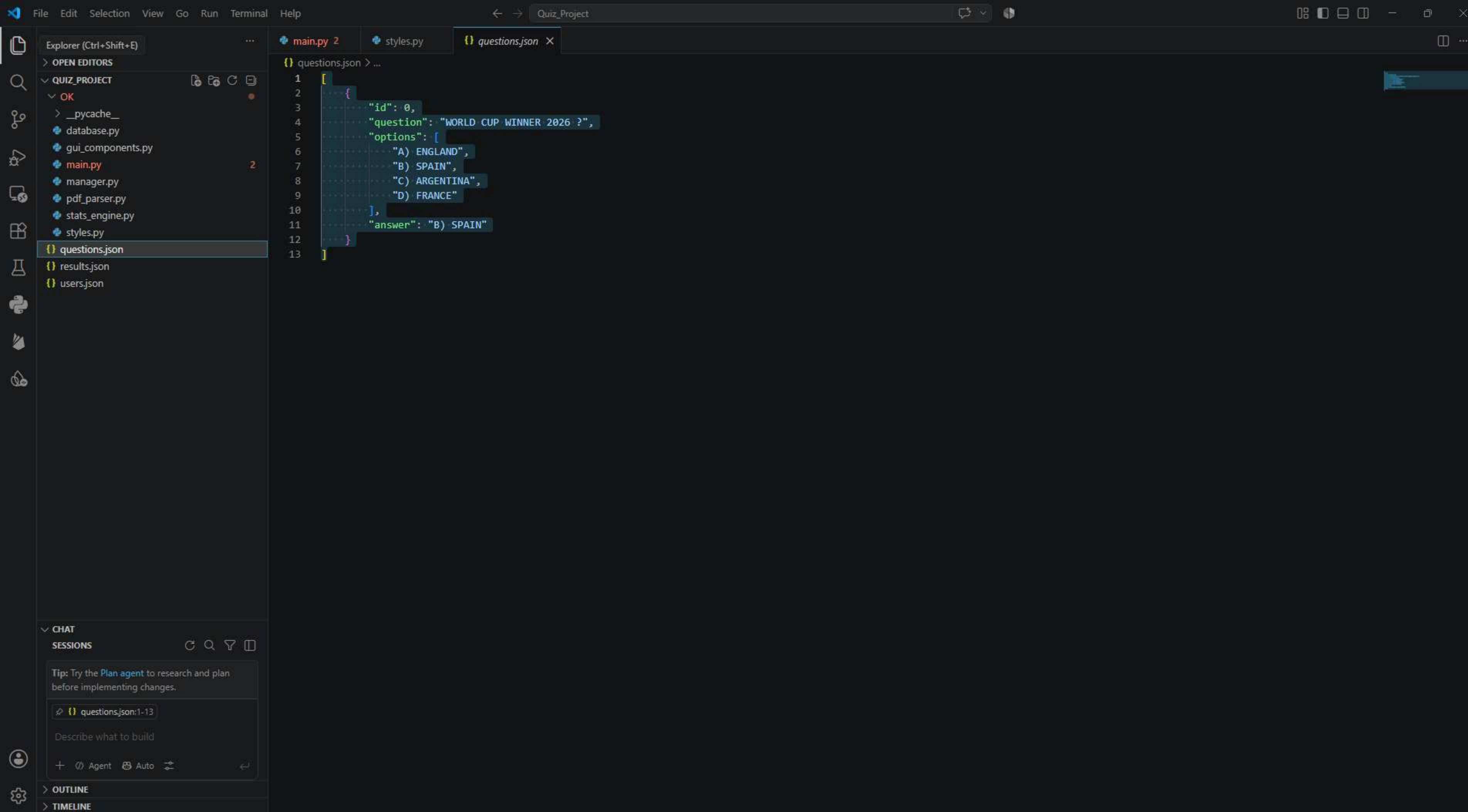
styles.py:9-11

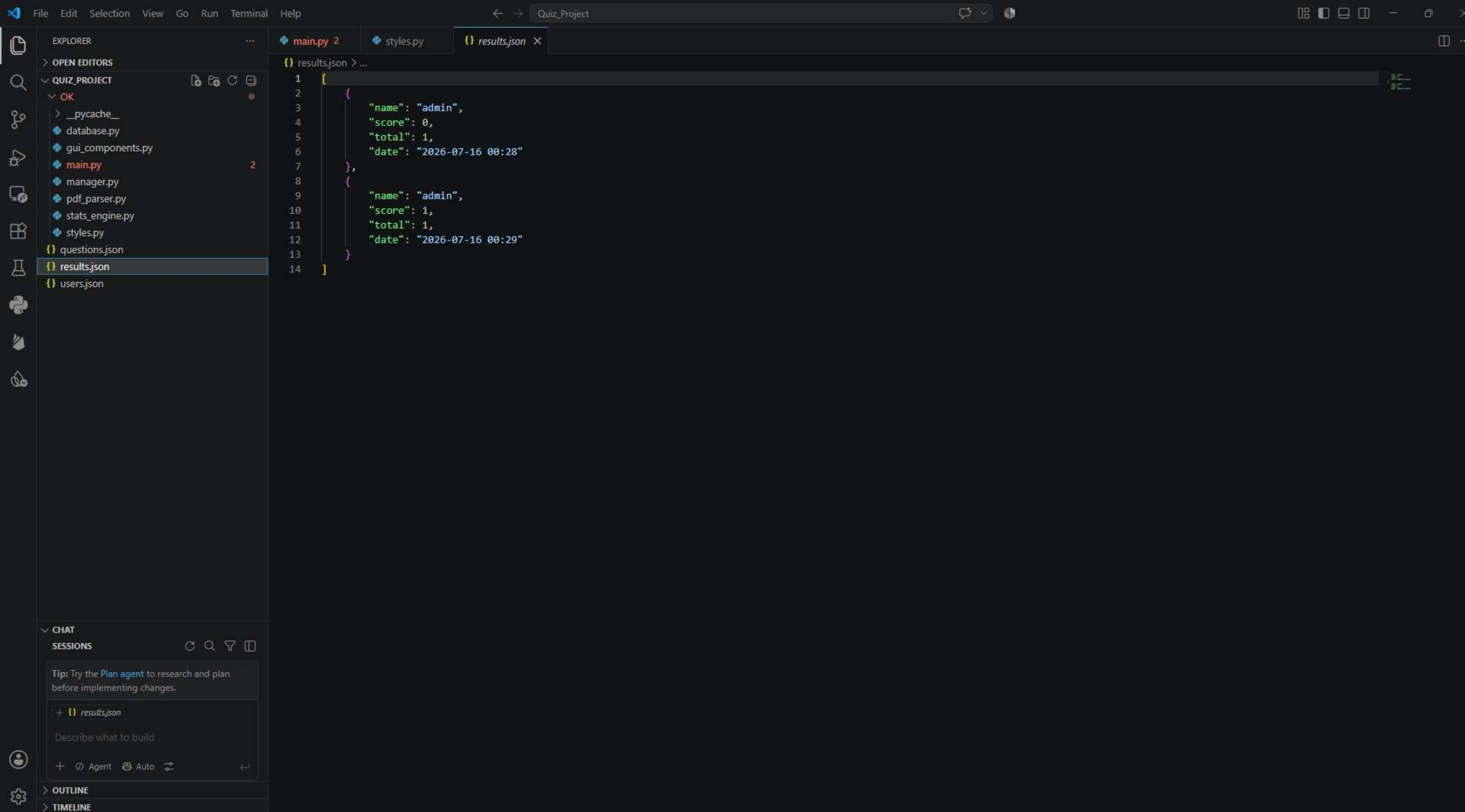
Describe what to build

+ Agent Auto

OUTLINE

TIMELINE





STUDENT LOGIN

Access quiz engine & analytics

Username

Password

Sign In

[New user? Register here](#)



UNIVERSITY

-  Home
-  Play Quiz
-  Question Manager
-  Leaderboard
-  Statistics
-  Save Local Data
-  Load Local Data
-  Logout
-  Exit

Welcome Back, ADMIN

Versity Quiz & Academic Result System

UNIVERSITY

QUESTION 1 OF 1

-  Home
-  Play Quiz
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WORLD CUP WINNER 2026 ?

- ☒ A) ENGLAND
- ☐ B) SPAIN
- ☐ C) ARGENTINA
- ☐ D) FRANCE

SUBMIT & NEXT →